Practical 1 — Categorical business data

Historical mutual funds · 307312 Applied Statistics — Masters Level

- This practical accompanies categorical variables, frequency tables, contingency tables and qualitative charts in Part 2.
- Complete Part 1 first.
- Allow **two sessions of approximately 40–50 minutes**: Sections 1–4, then Sections 5–8.

By the end, you should be able to:

- Summarize nominal and ordinal variables.
- Choose a suitable chart.
- Distinguish overall from conditional percentages.
- Write a business finding with a clear denominator.

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1. The business question

An analyst is reviewing a historical list of **259 mutual funds**:

> - How is this list distributed across investment objectives and risk classifications?
> - Does the proportion carrying a sales charge differ across risk groups?

- One row represents **one listed fund entry**, not an investor, purchase or year.
- The data are supplied with Berenson, Levine and Krehbiel's *Basic Business Statistics: Concepts and Applications*, ninth edition, as **FUNDS2002**.
- They describe historical classifications and returns through 2001, rather than today's market.

Course variable	Meaning	Treatment
`fund`	Fund label	Identifier
`type`	Large-, mid- or small-cap stocks	Category for this analysis
`objective`	Growth or Value	Nominal category
`sales_charge`	Whether a sales charge is recorded, No or Yes	Nominal category; the book calls this `Fees`
`risk`	Very low, low, average, high, very high	Ordered category
`turnover`	Trading activity in the same five ordered levels	Ordered category

- **Sales charge and expense ratio are different variables.**

- A fund recorded as having no sales charge can still have operating expenses.
- The risk categories are classifications, not equally spaced numerical measurements.

2. Import and inspect

- Open `Applied-Statistics.Rproj`, then this notebook.
- Install `dplyr` and `ggplot2` once if needed, as introduced in Part 1.
- Run the cells in order.
- Download the CSV.

```
# Locate the shared data folder from the course root or notebook folder.
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
funds <- read.csv(file.path(data_dir, "practicals/funds2002.csv"),
                  stringsAsFactors = FALSE)
suppressPackageStartupMessages(library(dplyr))
source(file.path(data_dir, "..", "practicals", "_includes", "petra-plots.R"))

# Ordered factors keep cumulative tables and plots in the intended order.
risk_levels <- c("very low", "low", "average", "high", "very high")
funds$risk <- factor(funds$risk, levels = risk_levels, ordered = TRUE)
funds$turnover <- factor(funds$turnover, levels = risk_levels, ordered = TRUE)
c(Records = nrow(funds), Variables = ncol(funds))
```

```
 Records Variables
     259        13
```

```
colSums(is.na(funds))
```

```
                fund                 type            objective
                   0                    0                    0
assets_million_dollars        sales_charge    expense_ratio_pct
                   0                    0                    0
       return_2001_pct       return_3yr_pct       return_5yr_pct
                   0                    0                    0
            turnover                 risk      best_quarter_pct
                   0                    0                    0
    worst_quarter_pct
                   0
```

- `read.csv()` imports the records.
- `factor()` defines category order.
- `is.na()` and `colSums()` check missing cells, including unrecognized factor labels.
- There are 259 records, 13 variables and no missing cells after this conversion.

```
# Select only the fields needed for the categorical questions.
knitr::kable(head(funds[c("fund", "type", "objective", "sales_charge", "risk")]))
```

fund	type	objective	sales_charge	risk
ABN Amro Montag & Calswell Growth I	large cap	Growth	No	average
AIM Aggressive Growth A	small cap	Growth	Yes	high
Aim Basic Value A	large cap	Value	Yes	low
Aim Small Cap Growth A	small cap	Growth	Yes	high
AIM Summit I	large cap	Growth	Yes	high
AIM Value B	large cap	Growth	No	average

Explain: Why can we calculate a percentage of Growth funds, but not a meaningful mean of the words Growth and Value?

3. One categorical variable

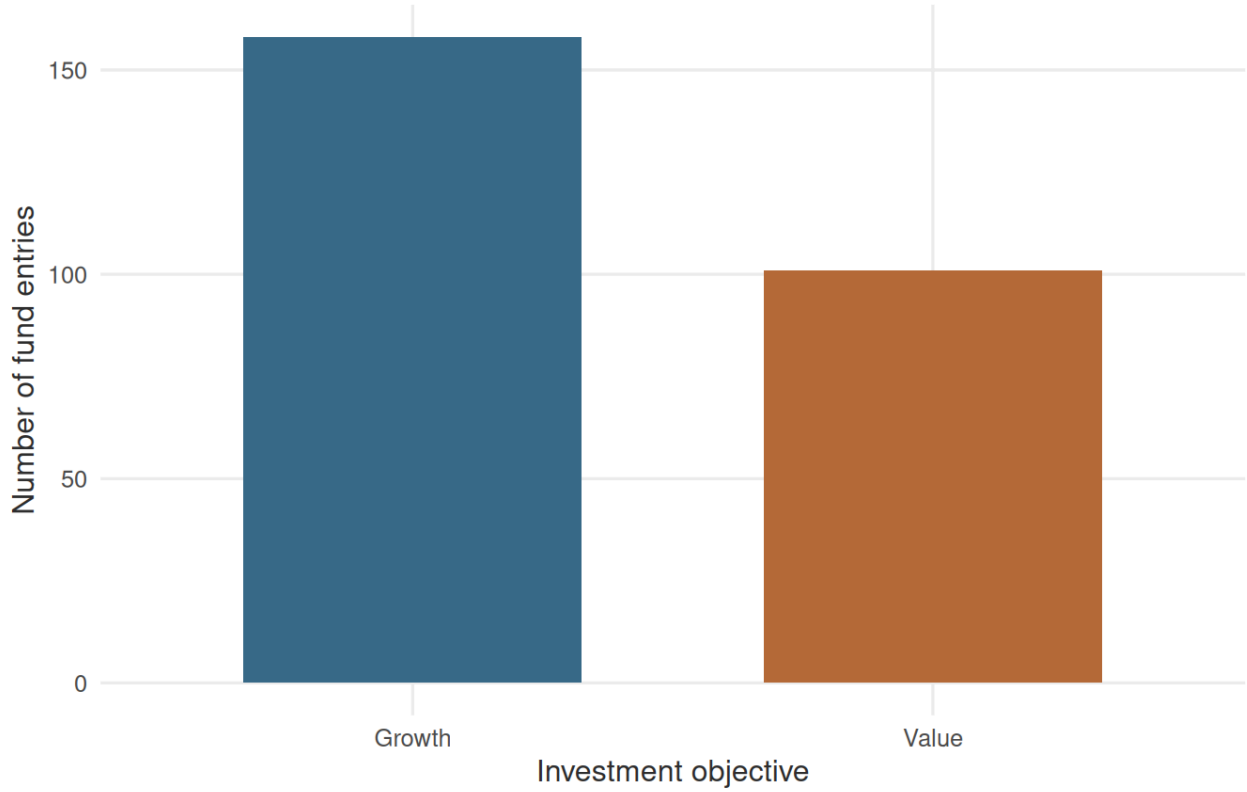
Question: What percentage of the listed funds have each investment objective?

```
# count() counts records; mutate() adds proportions and percentages.
objective_table <- funds |>
  count(objective, name = "Frequency") |>
  mutate(Relative_frequency = Frequency / sum(Frequency),
         Percentage = 100 * Relative_frequency)
knitr::kable(objective_table, digits = 3)
```

objective	Frequency	Relative_frequency	Percentage
Growth	158	0.61	61.004
Value	101	0.39	38.996

- There are **158 Growth funds (61.0%)** and **101 Value funds (39.0%)**.
- The denominator is all 259 entries in this list.
- This is not an estimate of the share of all funds in today's market.

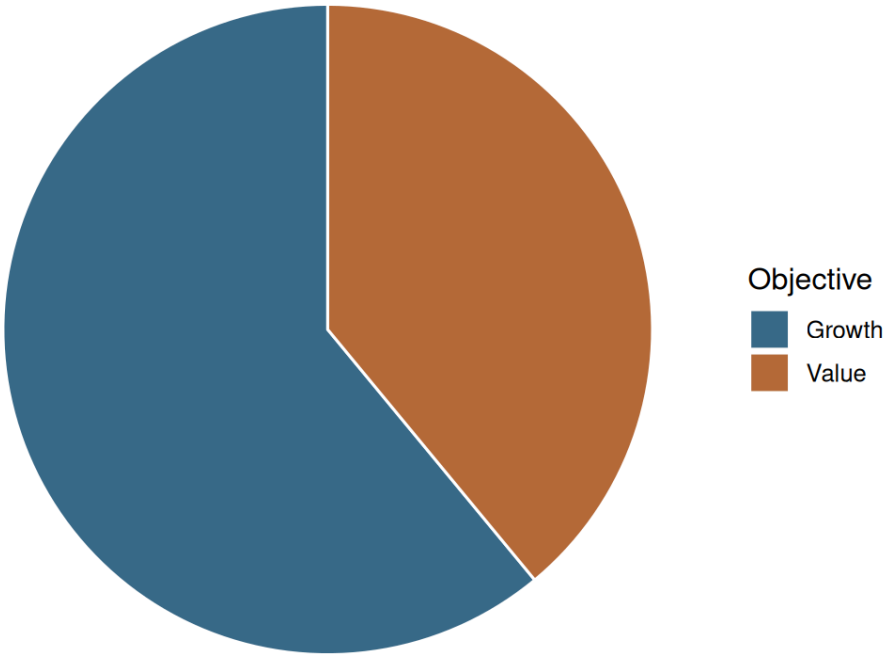
```
# geom_bar() counts records directly; use the same objective colors throughout.
ggplot(funds, aes(x = objective, fill = objective)) +
  geom_bar(width = 0.65) +
  scale_fill_manual(values = petra_groups, guide = "none") +
  labs(x = "Investment objective", y = "Number of fund entries")
```



Investment objective among the 259 historical fund entries.

- A bar chart makes comparisons easy.
- A pie chart can show parts of the same whole, but angles are harder to compare precisely.

```
# geom_col() uses computed counts; polar coordinates turn the stack into slices.
ggplot(objective_table, aes(x = "", y = Frequency, fill = objective)) +
  geom_col(width = 1, colour = "white") + coord_polar(theta = "y") +
  scale_fill_manual(values = petra_groups) +
  labs(x = NULL, y = NULL, fill = "Objective") + theme_void(base_size = 12)
```



Share of listed funds by objective; the two slices total 100%.

Your turn A

- Create a count-and-percentage table and a bar chart for `type`.
- Identify the largest category and state the denominator.
- Use the bar chart as your main comparison; the pie chart is an alternative display, not an additional required submission.

4. Ordered categories

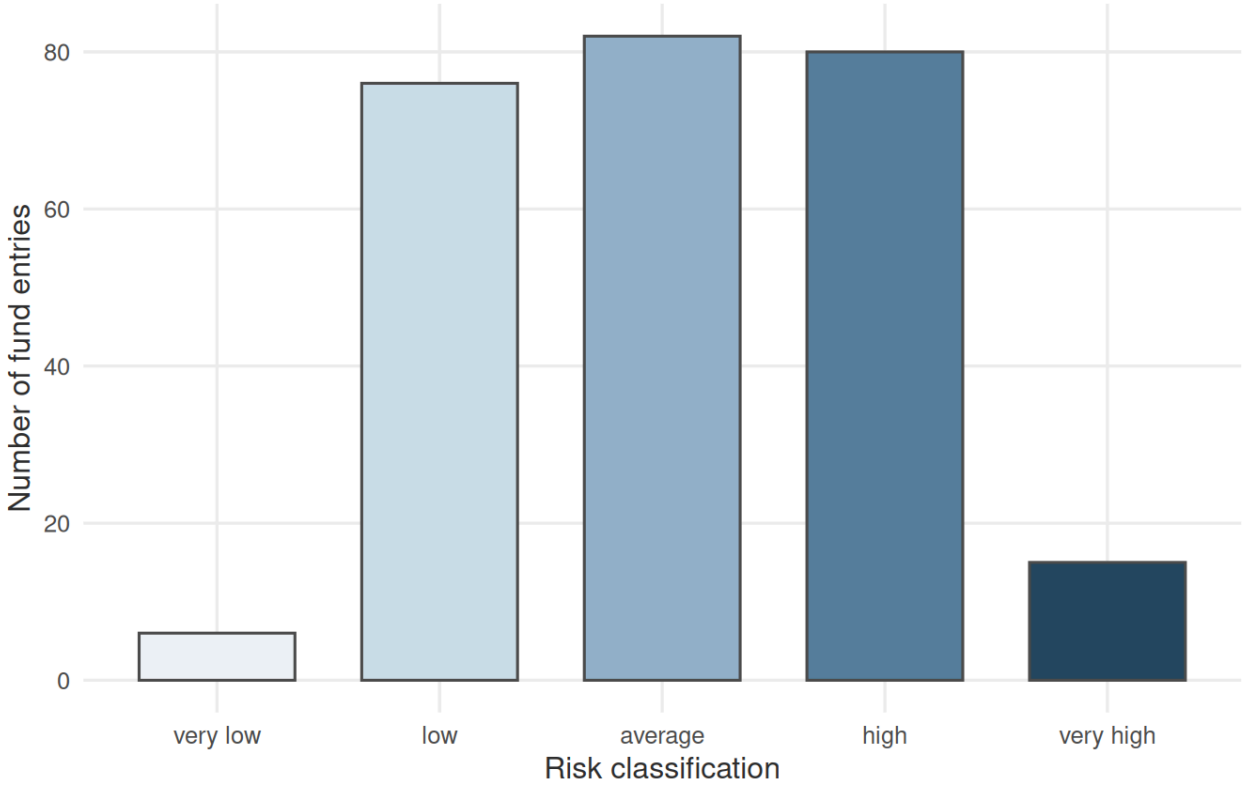
Question: What percentage of the listed funds have average risk or lower?

```
# .drop=FALSE retains every defined category; cumsum() calculates running totals.
risk_table <- funds |>
  count(risk, name = "Frequency", .drop = FALSE) |>
  mutate(Percentage = 100 * Frequency / sum(Frequency),
         Cumulative_frequency = cumsum(Frequency),
         Cumulative_percentage = 100 * Cumulative_frequency / sum(Frequency))
knitr::kable(risk_table, digits = 1)
```

risk	Frequency	Percentage	Cumulative_frequency	Cumulative_percentage
very low	6	2.3	6	2.3
low	76	29.3	82	31.7
average	82	31.7	164	63.3
high	80	30.9	244	94.2
very high	15	5.8	259	100.0

- The first three categories contain **164 / 259 = 63.3%** of the listed funds.
- That is average risk **or lower**.
- The Average category alone contains **82 / 259 = 31.7%**.

```
ggplot(risk_table, aes(x = risk, y = Frequency, fill = risk)) +
  geom_col(width = 0.7, colour = "#4B4B4B") +
  scale_fill_manual(values = petra_risk, guide = "none") +
  labs(x = "Risk classification", y = "Number of fund entries")
```



Risk classifications in their stated order.

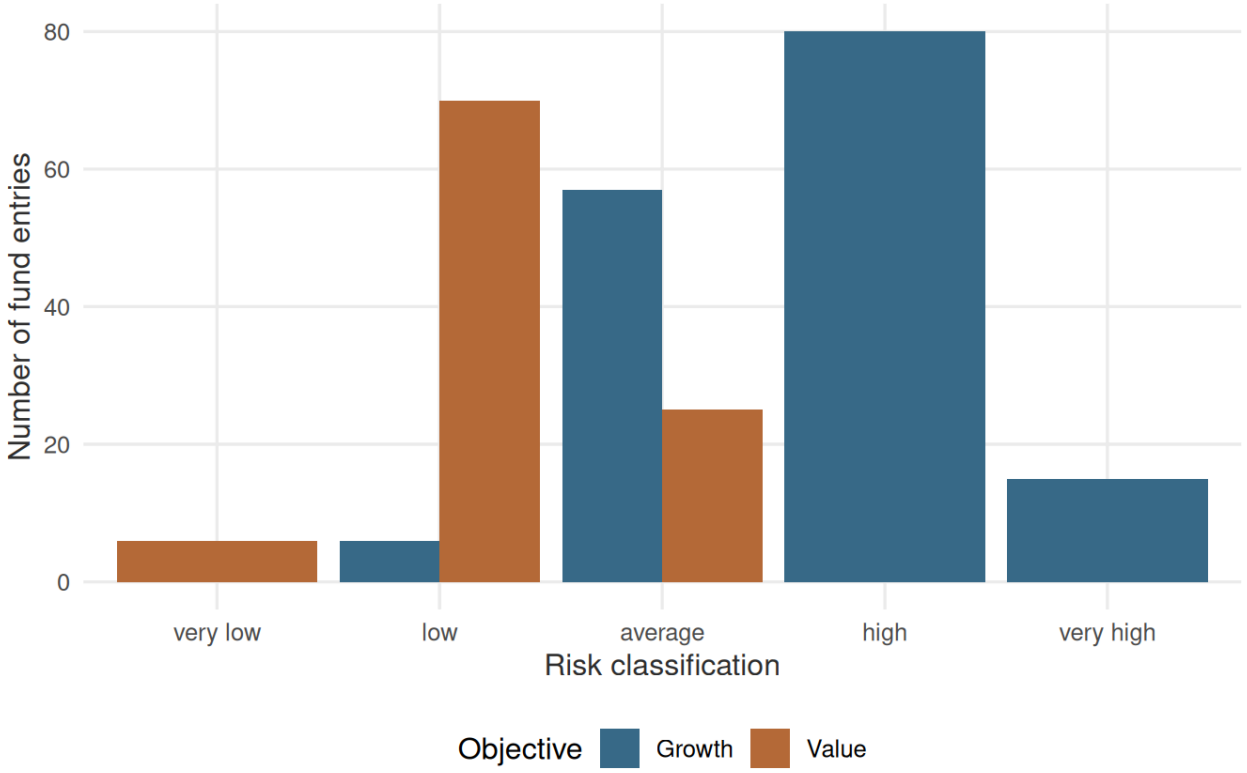
Your turn B

- Repeat the cumulative table for `turnover`.
- What percentage have low turnover or less?
- Explain why the ordering must be defined before calculating a running total.
- A cumulative percentage across Growth and Value would depend on an arbitrary ordering and has no useful interpretation here.

5. Compare categorical distributions

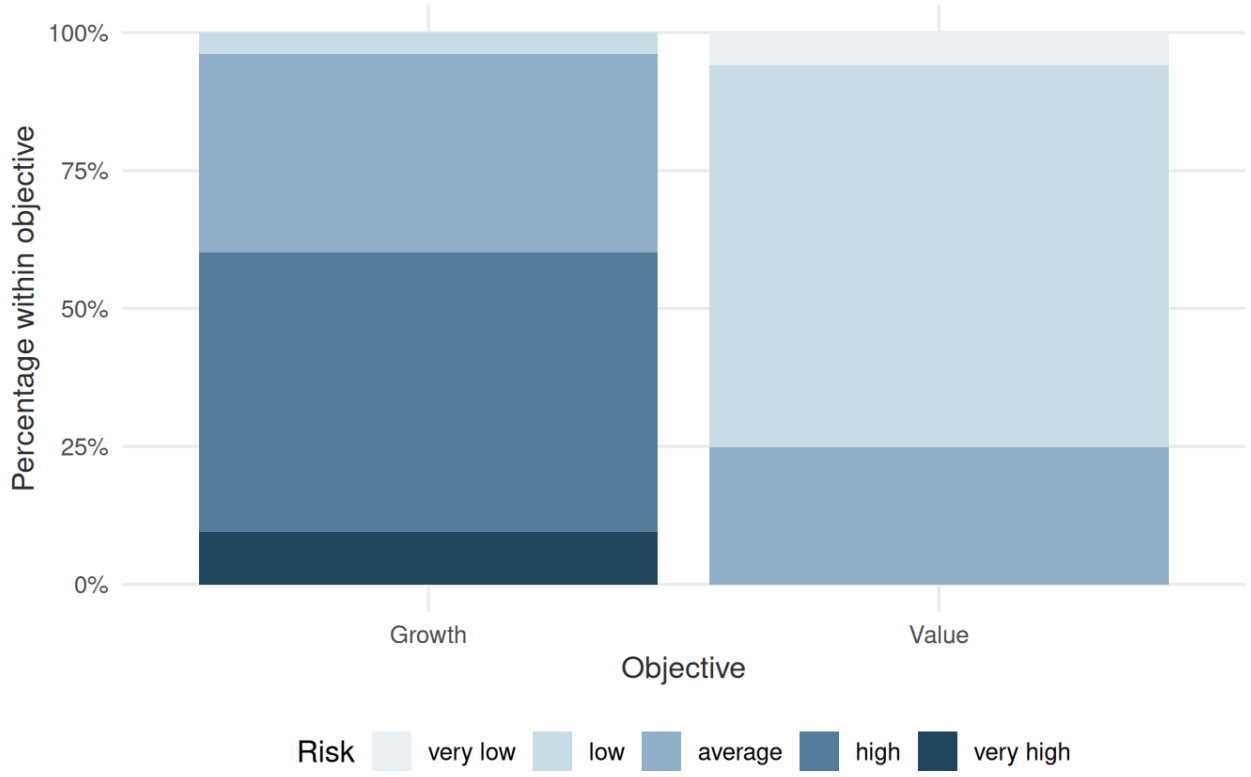
Question: How does risk composition differ between Growth and Value funds?

```
# position="dodge" places objective counts beside one another within each risk
        level.
ggplot(funds, aes(x = risk, fill = objective)) +
  geom_bar(position = "dodge") +
  scale_fill_manual(values = petra_groups) +
  labs(x = "Risk classification", y = "Number of fund entries", fill =
        "Objective")
```



- The groups contain different numbers of funds.
- Counts answer how many entries appear; percentages within each objective answer how each group's risk composition is distributed.

```
# position="fill" divides each stack by that objective's total.
ggplot(funds, aes(x = objective, fill = risk)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = petra_risk) +
  scale_y_continuous(labels = function(x) paste0(round(100 * x), "%")) +
  labs(x = "Objective", y = "Percentage within objective", fill = "Risk")
```



Risk distribution within each objective; each bar totals 100%.

Your turn C

- Compare sales-charge status across capitalization types using counts and a 100% stacked chart.
- Use `sales_charge`, rather than the numerical `expense_ratio_pct`.
- Describe how the two displays answer different questions.

6. Contingency tables and denominators

Question: Within each risk group, what proportion of funds have a sales charge?

```
# table() cross-classifies two variables; addmargins() adds count totals.
charge_risk <- table(Risk = funds$risk, Sales_charge = funds$sales_charge)
knitr::kable(addmargins(charge_risk))
```

	No	Yes	Sum
very low	4	2	6
low	45	31	76
average	59	23	82
high	45	35	80
very high	11	4	15
Sum	164	95	259

- There are 95 Yes and 164 No entries overall.

- Among the 80 High-risk funds, 35 have a sales charge.
- These values support three different statements:

Calculation	Question	Percentage
35 / 259	What share of all entries are both High risk and Yes?	13.5%
35 / 80	Within High risk, what share are Yes?	43.8%
35 / 95	Within Yes, what share are High risk?	36.8%

```
# No margin uses the grand total; margin 1 uses row totals; margin 2 uses column
        totals.
knitr::kable(round(100 * prop.table(charge_risk), 1), caption = "Overall
        percentages")
```

Overall percentages

	No	Yes
very low	1.5	0.8
low	17.4	12.0
average	22.8	8.9
high	17.4	13.5
very high	4.2	1.5

```
knitr::kable(round(100 * prop.table(charge_risk, margin = 1), 1), caption = "Row
        percentages")
```

Row percentages

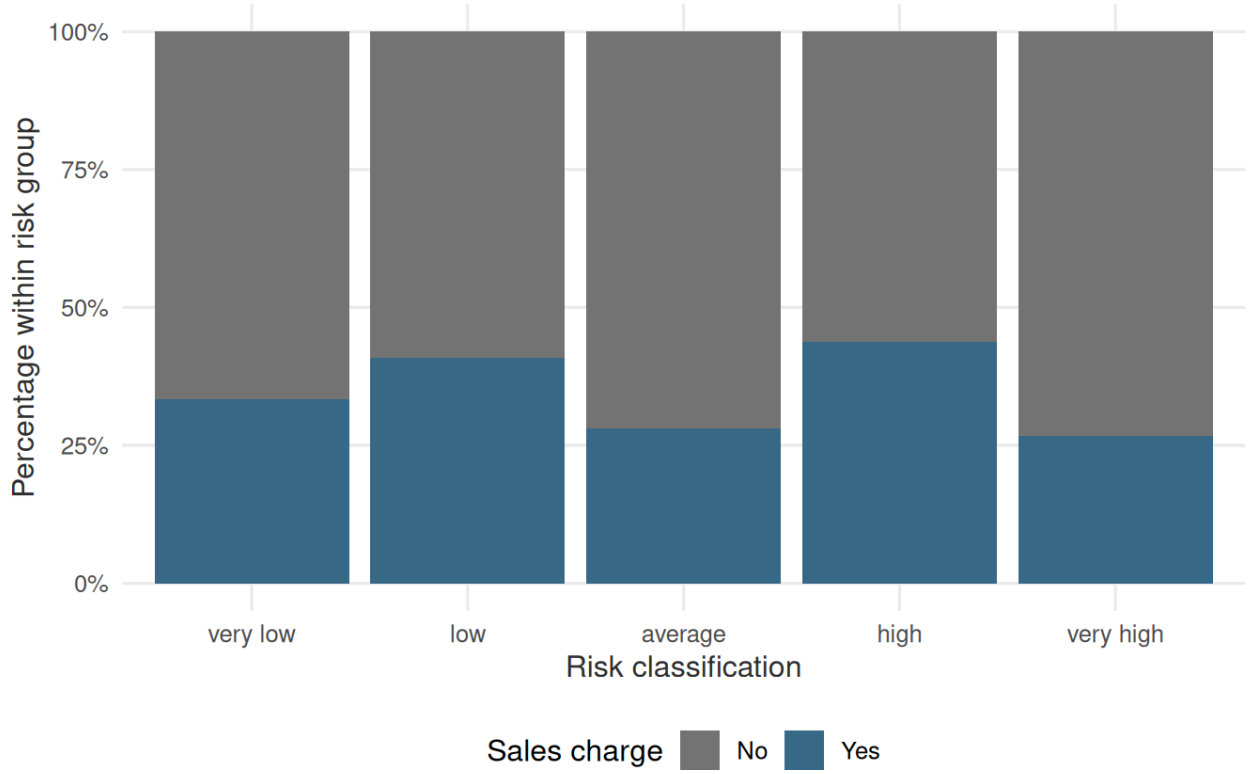
	No	Yes
very low	66.7	33.3
low	59.2	40.8
average	72.0	28.0
high	56.2	43.8
very high	73.3	26.7

```
knitr::kable(round(100 * prop.table(charge_risk, margin = 2), 1), caption =
        "Column percentages")
```

Column percentages

	No	Yes
very low	2.4	2.1
low	27.4	32.6
average	36.0	24.2
high	27.4	36.8
very high	6.7	4.2

```
ggplot(funds, aes(x = risk, fill = sales_charge)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = petra_charge) +
  scale_y_continuous(labels = function(x) paste0(round(100 * x), "%")) +
  labs(x = "Risk classification", y = "Percentage within risk group", fill =
       "Sales charge")
```



Sales-charge status within each risk group; equivalent to row percentages.

- The observed percentages describe this list.
- They do not establish that risk causes charges, or that any observed difference extends to a wider population.

Your turn D

- Construct `type` by `objective` counts and row percentages.
- Write one business question answered by the row percentages, then explain how the question changes if you use column percentages.

7. Interpretation workshop

Decide whether each claim is supported, and correct any misleading statement.

1. Most entries in this list have a Growth objective.
2. 63.3% of the funds are classified as Average risk.
3. 43.8% of all entries are High-risk funds with a sales charge.
4. A fund with no sales charge has no expenses.
5. Risk classifications explain why a fund charges a sales fee.

For each reported percentage, identify the numerator, denominator and group being described.

8. Independent briefing

Prepare a concise briefing for the analyst with:

1. One table describing capitalization types and their percentages.
2. One ordered cumulative table for turnover.
3. One chart comparing sales-charge percentages within capitalization types.
4. Three findings with explicit denominators and one limitation.

- Submit your completed Quarto notebook and rendered HTML.
- Keep the briefing to approximately **one page**, with supporting code and output below it.
- Credit depends on method choice and interpretation as well as correct calculations.

Source and data notes

- Berenson, Levine and Krehbiel, *Basic Business Statistics: Concepts and Applications*, ninth edition, **FUNDS2002** companion data.
- The local ninth-edition book defines the fund comparison on p. 40 and variables on p. 75 (team project TP2.1); page numbers differ from the international edition cited by the supplied draft.
- Category ordering and the guided tasks here support the course's Part 2 sequence.
- The CSV preserves source observations and order; headers were renamed and surrounding spaces in `Type` were trimmed.
- The underlying collection and probability-selection method have not been established, so conclusions concern the supplied list.
- Data dictionary and provenance.

Continue with Practical 2: Numerical business data.